AZATHIOPRINE

Check TPMT before starting

Allopurinol interaction

CHEMICAL NAME 6-[(1-methyl-4-nitro-1H-imidazol-5-yl)sulfanyl]-7H-purine

CLASS Antimetabolite immunosuppressant (purine analogue, prodrug of 6-mercaptopurine)

TRADE NAMES Imuran · Azasan · Azamun

THERAPEUTIC USES / INDICATIONS

- Prevention of organ transplant rejection.
- Inflammatory bowel disease (steroid-sparing maintenance).
- Autoimmune diseases: SLE, vasculitis, autoimmune hepatitis, myasthenia gravis, rheumatoid arthritis, dermatological conditions.

ADMINISTRATION

- **PO** with or after food (↓ nausea), once daily or divided.
- **IV** (rarely) — irritant.
- Cytotoxic handling precautions.

DOSAGES

- **Autoimmune/IBD:** 1–2.5 mg/kg/day (start 50 mg, increase gradually).
- **Transplant:** 1–3 mg/kg/day.
- TPMT activity: low → reduced dose (e.g. 10–50%); absent → avoid.
- With allopurinol: reduce to 25% of usual dose.

PHARMACOKINETICS

- **Onset:** 6–12 weeks for autoimmune effect.
- Converted to 6-mercaptopurine, then thiopurine nucleotides; inactivated by TPMT and xanthine oxidase.
- Excreted in urine. **t½:** 5 h (active metabolites longer).

PHARMACODYNAMICS

- Purine antagonist → ↓ DNA/RNA synthesis in proliferating T and B lymphocytes → immunosuppression.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, anorexia.
- Rash, hair loss.
- Flu-like symptoms.

ADVERSE EFFECTS (SERIOUS)

- Bone-marrow suppression (leucopenia, thrombocytopenia); severe in TPMT deficiency.
- Hepatotoxicity (cholestasis, veno-occlusive disease).
- Pancreatitis.
- Infections; malignancy (skin cancer, lymphoma) long-term.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to azathioprine or mercaptopurine.
- Absent TPMT activity.
- Live vaccines; pregnancy (relative — often continued under specialist).
- Severe infections.

KEY INTERACTIONS

- Allopurinol, febuxostat (xanthine oxidase inhibitors): ↑↑ toxicity; reduce dose to 25% or avoid.
- **Warfarin:** ↓ anticoagulant effect.
- **Co-trimoxazole, ACE inhibitors:** ↑ myelosuppression.

NURSING MANAGEMENT

- Check TPMT result before first dose.
- FBC and LFTs weekly for first 4–8 weeks, then at least every 3 months.
- Ask about sore throat, fever, bruising, abdominal pain.
- Check for allopurinol.
- Sun protection advice.

PATIENT EDUCATION

- Take after food.
- Report infections, bruising, bleeding, severe abdominal pain, yellow skin.
- Use sunscreen and check skin regularly (skin cancer risk).
- Avoid live vaccines.

KEY POINTS

- Purine analogue immunosuppressant.
- Marrow suppression, pancreatitis, hepatotoxicity.
- Check TPMT; 1–2.5 mg/kg/day.
- Allopurinol → reduce dose to 25%.

SOURCES CHECKED **[849]** Azathioprine 25mg Film-Coated Tablets (UK SmPC) **[850]** Imuran, oral (US label) **[851]** Azathioprine, oral (US label) *Full references p. 432–457*